CNN / Siamese Embedding Matching — Phase A Results

Contactless Fingerprint Matching SDK (UIDAI SITAA, project SU-C1-2025-02) — off-the-shelf pretrained CNN embedding baseline

1. Objective

The deployed MinutiaNet-based matcher (minutiae extraction + geometric matching, backend/matching.py) measures ~52% accuracy on contactless fingerprints. This experiment tests whether comparing deep CNN image embeddings via cosine similarity — instead of comparing extracted minutiae points — provides better or complementary matching performance, per the team's own next-plan writeup.

2. Method (Phase A: pretrained, no training)

Each fingerprint image was passed through an ImageNet-pretrained ResNet18 with its final classification layer removed, producing a 512-dimensional embedding vector per image. No fingerprint-specific training was performed — this is a fixed, off-the-shelf feature extractor used purely to test whether generic deep image features already separate genuine from impostor fingerprint pairs, before investing in training a fingerprint-specific Siamese network (Phase B, not run here). Two embeddings were compared via cosine similarity, producing a single similarity score per pair.

3. Dataset and evaluation protocol

Source	HKPU Contactless 2D Fingerprint Database, first_session
Unique identities	336 (verified from the dataset's own folder count)
Images per identity	6 (1 enrolled as gallery, 5 held out as probes)
Gallery images	336
Probe images	1,680
Genuine pairs (probe vs. own gallery image)	1,680
Impostor pairs (probe vs. every other gallery image)	562,800

Each of the 336 identities was verified as a 1:1 genuine-vs-impostor test using the dataset's own ground-truth folder labels (each pX folder is a physically distinct finger photographed 6 times by the dataset's creators) — not a blind 1:N search. This mirrors the FAR/FRR/EER evaluation already used by backend/benchmark.py for the minutiae matcher.

4. Score distributions

	Mean	Median	Std. Dev.
Genuine pairs (same finger)	0.9741	0.9800	0.0210

Impostor pairs (different finger)	0.8862	0.8942	0.0452

5. Headline results

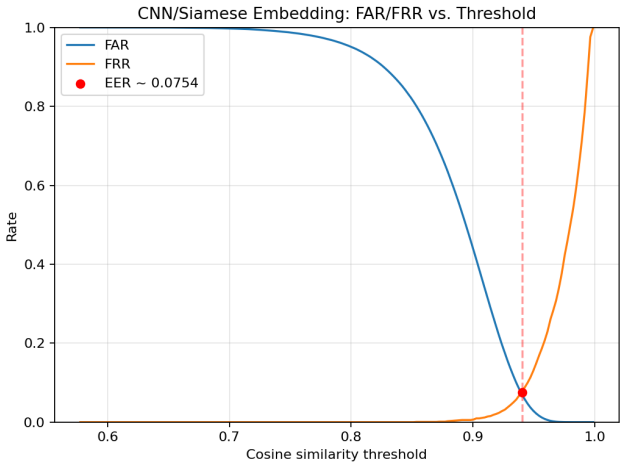
Metric	At EER threshold (≈ 0.940)	At illustrative threshold (0.75)
Matching Accuracy	92.46%	50.65%
FAR (False Accept Rate)	7.05%	98.69%
FRR (False Reject Rate)	8.04%	0.00%
EER (Equal Error Rate)	7.54%	—

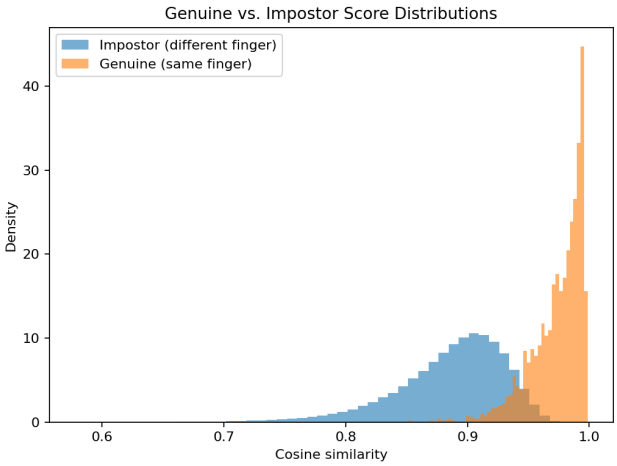
Note on threshold choice: the PDF's own illustrative threshold of 0.75 performs poorly here (50.7% accuracy) because this embedding space clusters much higher than that example assumed — genuine and impostor scores both sit mostly in the 0.85–0.99 range for this generic, same-domain (all fingerprints) network. The threshold must be calibrated to the actual score distribution (≈ 0.94 here), not assumed from an unrelated example.

6. Baseline comparison

Method	Matching Accuracy
MinutiaNet baseline (existing pipeline)	52.0%
CNN/Siamese embedding — Phase A (this report)	92.46%

7. FAR/FRR curve and score distributions





8. Scope and caveats

• **Phase A only.** The ResNet18 used here was never trained or fine-tuned on any fingerprint data — it is a generic, off-the-shelf ImageNet feature extractor. This result shows deep embeddings carry real discriminative signal for this problem; it is not yet the result of a fingerprint-specific Siamese network.

• **1:1 verification, not 1:N identification.** Every pair scored here has a known ground-truth label from the dataset's own folder structure (same pX folder = same finger). This measures "does the similarity score correctly say match/no-match for a known pair," not "can the system correctly pick the right person out of all 336 given an unlabeled probe" (rank-1 identification accuracy) — that is a separate, stricter test not run in this pass.

• **Single-session only.** Only first_session images were used (336 unique identities). second_session's 160 people are recaptures of a subset of these same identities months later, per the dataset's own readme, and were excluded from this pass rather than double-counted as new identities.

• **Not yet a hybrid or fine-tuned result.** The PDF's proposed next steps — fine-tuning with contrastive/triplet loss on a held-out identity split (Phase B), and combining minutiae + deep-embedding scores (hybrid matching) — are not covered by this report.

Generated from siamese_embedding/evaluate.py output (benchmark_results.csv, eer_plot.png) on the siamese-embedding branch.